

Declarations

Conflict of Interest

The author declares no competing interests.

Funding

This research received no external funding.

Ethics

This study used publicly available de-identified datasets; therefore, ethical approval was not required.

Data Availability

Both datasets used in this study are publicly available. The ISIC 2018 dataset is accessible through the ISIC Archive (<https://challenge.isic-archive.com/data/>). The PAD-UFES-20 dataset is available at <https://data.mendeley.com/datasets/zr7vgbcyr2>. Code will be made publicly available upon acceptance.

AI Usage Statement

During the preparation of this work, the author(s) used Claude (Anthropic) to assist with statistical analysis implementation, figure generation, and drafting of manuscript text. All research questions, experimental design decisions, interpretation of results, and scientific conclusions were determined by the author(s). The author(s) reviewed, verified, and edited all AI-assisted content and take full responsibility for the content of this publication.

Author Contributions

Duc-Vinh Tran: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Visualization, Writing – original draft, Writing – review & editing.